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The Evolution and Impact of Data Science in the National Basketball Association

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This is to certify that, except where specific reference is made, the work described in this project is the result of the investigation carried out by the student, and that neither this project nor any part of it has been presented, or is currently being submitted in candidature for any award other than in part for the MSc Data Science degree of the University of South Wales.

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Abstract

Basketball has undergone a significant transformation over the past two decades, driven by the integration of advanced analytics into decision-making, talent evaluation, and tactical design. The rise of the three-point shot, player-tracking technologies, and predictive modelling frameworks has reshaped how teams approach efficiency and success. This dissertation investigates the impact of data science on the evolution of basketball strategy, with a particular focus on shot selection, offensive efficiency, and the redistribution of responsibility between star players and role players. Using season-level NBA datasets, forecasting models such as ARIMA, SARIMAX, and Prophet are employed to predict future trends in shot distribution and efficiency. Structural time-series models attribute observed shifts to identifiable league policies (e.g., SportVU adoption, emphasis on freedom of movement), while optimisation techniques quantify the gap between actual and theoretical efficiency frontiers. To extend the comparative scope, the research also applies analogous concepts to European football, examining whether usage concentration and player involvement metrics mirror basketball's star–role player dynamics. The findings highlight the explanatory power of analytics in linking policy interventions to tactical evolution, revealing that both basketball and football exhibit a measurable shift toward distributed offensive responsibility.

This work contributes to the growing body of applied sports analytics literature by combining predictive, causal, and optimisation-based approaches across two globally significant sports.

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Introduction

1.1 Background

In recent decades, professional sports have become increasingly shaped by the availability of rich datasets and the development of advanced statistical methods. Basketball, particularly the National Basketball Association (NBA), has emerged as a pioneer in the use of analytics, leveraging data science to influence shot selection, roster construction, and game strategy. The widespread adoption of player-tracking technologies, along with the emerging popularity of metrics such as effective field goal percentage and usage rate, has altered the way teams evaluate efficiency and distribute offensive responsibility.

Parallel to basketball's analytical revolution, european football has also begun to embrace performance analytics, with frameworks such as expected goals (xG) and expected assists (xA) informing tactical planning and recruitment. Although football's adoption has been uneven compared to basketball's, both sports now operate within data-informed ecosystems where evidence-based decision-making is increasingly integral to competitive success.

This dissertation positions itself within this growing field of applied sports analytics, aiming to assess how data science has influenced the tactical evolution of basketball and whether similar dynamics are observable in football.

1.2 Problem Statement

While basketball analytics have been extensively studied, especially in relation to the rise of the three-point shot and the efficiency revolution, there remains limited empirical work that links tactical changes directly to identifiable league policies and innovations such as player tracking. Moreover, although comparative analyses of basketball and football exist, few attempt to operationalise concepts such as "usage rate" across sports.

The central problem addressed by this research is therefore twofold:

To determine how analytics and league-level interventions have causally influenced the evolution of basketball strategies, and to assess whether analogous redistributions of offensive responsibility from star players towards role players can be observed in both sports.

1.3 Aims and Objectives

1.3.1 Aim

To evaluate the impact of data science and analytics on the evolution of offensive strategies in basketball and, to a lesser extent, football, with a focus on shot efficiency, forecasting, policy attribution, and player responsibility distribution.

1.3.2 Objectives

- To build and evaluate time-series forecasting models (ARIMA, SARIMAX, Prophet) predicting the trajectory of key offensive metrics such as three-point attempts per 100 possessions and effective field goal percentage.
- To apply structural time-series models to quantify the causal impact of identifiable policy interventions (e.g., SportVU adoption, rule changes) on league-wide tactical trends.
- To construct an efficiency frontier optimisation model that benchmarks actual shot distributions against theoretical optima under both risk-neutral and risk-aware conditions.
- To develop measures of star burden and role-player contribution in basketball, and to extend these metrics to football using player-season datasets.
- To compare patterns across basketball and football, assessing whether increased reliance on analytics corresponds to more distributed offensive responsibility.

1.4 Research Hypotheses

The study tests the following hypotheses:

H1: The adoption of analytics and player-tracking technology in the NBA is significantly associated with increases in three-point attempts and overall offensive efficiency.

H2: Forecasting models that incorporate exogenous variables (e.g., pace, offensive rebounding rate, free throw rate) outperform baseline autoregressive models in predicting tactical trends.

H3: Actual league-wide shot distributions fall below the classic efficiency frontier, as it disregards the weight of the three-pointer (1 more point per possession if drained) and focuses empirically on raw efficiency.

H4: In basketball, the reliance on high-usage stars has declined over the past decade, with role players contributing an increasing share of offensive actions.

H5: Football exhibits analogous patterns of redistributing creative responsibility, with lower usage concentration (HHI) associated with greater team success.

Literature Review / Related Work

The increasing integration of data science into professional sports has led to a profound shift in how games are played, analysed, and perceived. Basketball, in particular, has experienced a data-driven transformation, with teams, coaches, and analysts alike using advanced metrics to optimise performance and guide decision-making. This literature review explores related academic work and industry findings on how analytics has shaped modern basketball.

2.1 The Rise of Basketball Analytics

A study in the *Journal of Sports Economics* (Wang 2025) quantified this trend by linking NBA teams' investments in analytics staff to on-court success (Chu 2025). The authors found that for roughly every 0.8 addition to a team's analytics headcount, the team gained one extra win per season, a contribution valued equivalently to about $9.6 million in player salary (Wang 2025). In other words, the impact of a data analyst was estimated to be on par with a mid-level player contract, which is a striking indication of how much analytics have become embedded in team strategy and roster planning. Notably, as of the mid-2020s, every NBA franchise employs at least one dedicated analytics staffer (Wang 2025), whereas two decades ago, such roles were virtually nonexistent, underscoring what is an industry-wide shift where evidence-based decision-making is now seen as a competitive necessity rather than a luxury.

Analytics-driven insights have influenced various aspects of the game. Early pioneers like Dean Oliver(Wikipedia contributors 2025) (often considered basketball's Bill James) introduced advanced metrics (e.g. offensive/defensive ratings and the "Four Factors") that demonstrated better correlations with winning than traditional box scores (Taylor & Bechtold 2020). In the 2000s, teams slowly began to embrace these ideas, and by the 2010s, the "Moneyball"-inspired mindset had fully arrived in the NBA (Chu 2025). For instance, teams such as the Houston Rockets, under Daryl Morey's management, famously reoriented their strategy around efficient shot selection, a move that was entirely driven by statistical findings. Morey's analysis showed that shooting more three-pointers wins more games, which directly challenged and changed conventional coaching wisdom.

Figure 2, derived from NBAstuffer data (Wang 2025), illustrates this paradigm shift by showing the steady increase in NBA teams' analytics staff sizes over time, reflecting teams' growing commitment to data-driven approaches (with missed data in 2011 and 2018 due to lockout-related factors). Collectively, these developments chart the rise of basketball analytics from early adoption to its current status as a cornerstone of team operations.

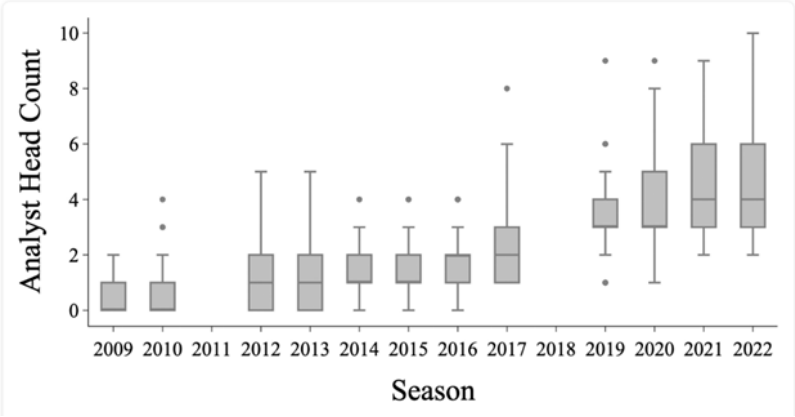

Figure 2.1: Box plots of NBA analytics department headcounts over time Wang (2025).

2.1.1 Shot Efficiency and the Three-Point Revolution

One of the most significant transformations in modern basketball is the rise of the three-point shot, a trend that has reshaped offensive philosophies across the NBA and other leagues. This phenomenon, which is often referred to as the "three-point revolution", is rooted in the pursuit of shot efficiency, as with a wider adoption of statistical analysis, teams started to realise that the expected value of a three-point attempt (due to its higher point yield) often outweighs that of a mid-range two-pointer, despite the lower conversion percentage. The concept of effective field goal percentage (eFG%), which weights three-pointers appropriately, became central to this evaluation. This metric allows teams to quantify shooting efficiency by accounting for the extra point a three-pointer provides, offering a clearer comparison across players and shot types.

As highlighted by Kilcoyne's analysis (Kilcoyne 2020), the adoption of eFG% in player evaluations has redefined how scouts and front offices assess shooting value, often revealing that players with strong three-point and at-rim scoring can outperform those who rely heavily on long two-point jumpers, even if their raw field-goal percentage appears similar.

In practice, NBA teams in the late 1990s and early 2000s averaged fewer than 15 three-point attempts per game; by the 2022–2023 season, that figure had skyrocketed to over 34 attempts per game.

This dramatic increase (illustrated in Figure 3) was not by accident but driven by analytical models optimising expected points per shot, with the strategic shift that was exemplified by teams like the Golden State Warriors and Houston Rockets, who prioritised high-value shots (three-pointers and efficient close-range shots) over the once-prevalent mid-range jumper, doing so while in the midst of historical regular season success.

As mentioned earlier, Figure 3 shows how shot location distributions changed league-wide: mid-range attempts declined sharply, while three-point attempts (especially from the corners and above the break) and shots at the rim rose in frequency. This reallocation of shots has led to record-high offensive efficiency in recent seasons, validating the data-driven approach to shot selection.

Academic research supports these trends. A study by Caporale and Collier (Caporale 2015) found that NBA teams that optimised shot selection (emphasising threes and efficient twos) tended to have better offensive ratings and win outcomes.

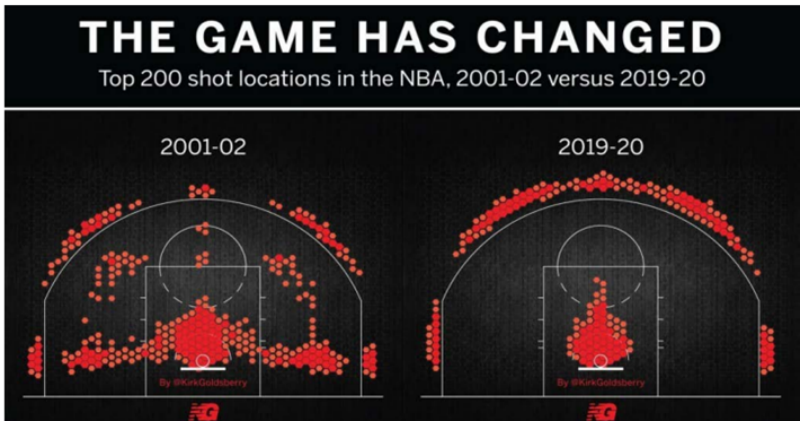

Figure 2.2: Shift in shot locations, driven by analytical efficiency Kilcoyne (2020).

Another analysis by Sliz (Sliz 2016) leveraged player-tracking data to examine conditions under which three-point shots are most successful. Sliz found that the best predictors of a three-point shot's success were:(1) the distance of the closest defender, open shots greatly improve success rates, (2) the speed of the catch-and-shoot release, as a quick release gives defences less time to contest, and (3) the ball movement leading up to the shot, shots off of multiple passes (indicating defensive rotation and possible breakdowns) yielded higher accuracy. All of these factors are measurable via modern data and underscore principles that coaches now emphasise (e.g., spacing the floor to increase defender distance, and "good to great" passing to find the truly open shooter).

The three-point revolution exemplifies how data science has altered basketball strategy at a fundamental level. It also serves as a template for other sports and leagues: identify the most efficient plays or actions and adjust tactics to maximise those opportunities.

2.1.2 Player Tracking and Spatial Analytics

Since the NBA introduced SportVU cameras for player tracking in 2013 National Basketball Association (2016), the richness of player data has grown exponentially. These systems capture the movement of all players and the ball 25 times per second, producing a data set prepared for spatial analytics.

One application of tracking data is the analysis of shot trajectories and probabilities. For example, Rajiv Shah et al. (2016) applied deep learning (recurrent neural networks) to SportVU trajectory data to predict whether a given three-point shot will go in or not (Shah & Romijnders 2016). Their RNN models, which considered the ball's motion and angle, outperformed traditional logistic regression models, demonstrating the value of neural networks in extracting subtle patterns from movement sequences. Figure 2.3 illustrates a related concept, namely physics-based free-throw shot predictions, highlighting how factors like launch angle and rotation can be modelled to assess the success of a shot.

Another major benefit of tracking data is the ability to evaluate defensive performance and off-ball player movement, which were once considered nearly unquantifiable. By analysing millions of position coordinates, metrics such as defensive coverage area, average defender distance, and

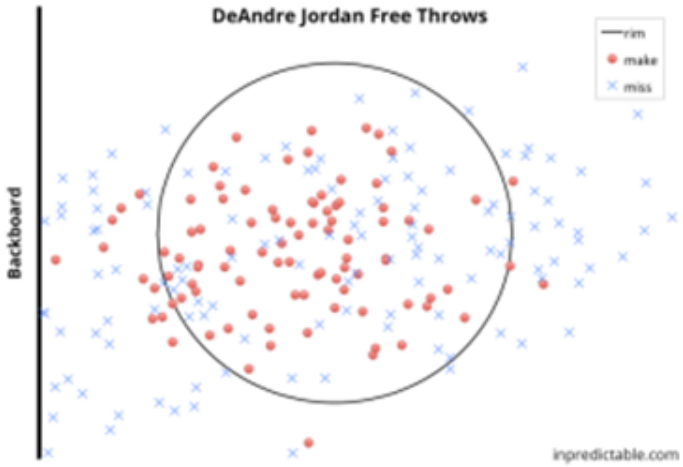

Figure 2.3: Physics-based free throw shot predictions Shah & Romijnders (2016).

off-ball screen effectiveness have been developed. For instance, one can compute how effectively a defender restricts an opponent's space or how often an offensive player successfully cuts behind the defence for an open look. These insights feed into coaching decisions: lineups can be optimised for defensive synergy (by looking at spatial help patterns), and playbooks can be adjusted to exploit opponents' spatial weaknesses.

Moreover, modern data visualisation tools enable teams to turn tracking data into intuitive visuals. Using Python libraries (matplotlib, seaborn) in Jupyter Notebooks, analysts regularly produce heatmaps (e.g., shooting heatmaps for individual players or teams), shot charts showing make/miss clusters, and possession maps that trace player movements during set plays.

2.1.3 Draft Analytics and Talent Identification

Scouting has historically been qualitative, but modern front offices blend film review with machine learning evaluations. With structured player datasets (e.g., NBA Draft Combine Measurements), analysts build predictive models for professional success. Kannan et al. Kannan et al. (2018) developed a machine learning framework combining college stats, biometric data, and draft position to predict NBA career outcomes. They highlighted draft order and college statistics as the strongest predictors.

They also demonstrated that tree-based models such as random forest and XGBoost capture non-linear relationships between pre-draft stats and long-term NBA success. Using Basketball Reference and draft combine data, their models achieved over 80% classification accuracy in predicting whether a player would exceed 200 career games.

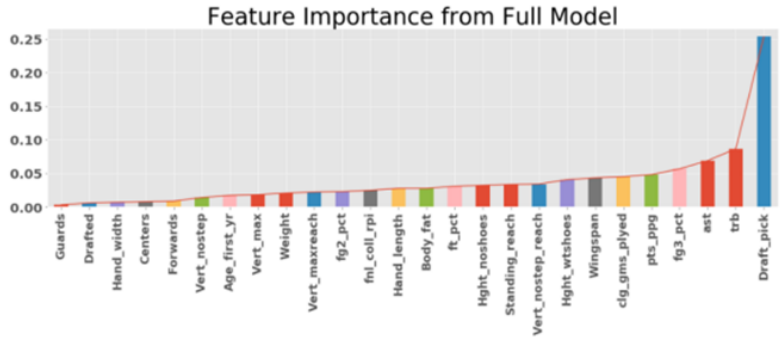

Figure 2.4: Feature importance values from draft success model Kannan et al. (2018).

2.1.4 Comparative Efficiency: NBA vs EuroLeague

While the NBA has become the global benchmark for basketball innovation and analytics, the EuroLeague (Europe's top club competition) offers an excellent counterpoint – a high-level league with different tactical philosophies, rules (e.g. a shorter three-point line), and resource constraints. Both leagues have embraced analytics to varying degrees, providing an opportunity to compare how data-driven efficiency manifests in different contexts.

A 2025 comparative analysis by Foteinakis and Pavlidou examined a decade of shooting trends and offensive efficiency in the NBA and EuroLeague (2012–2023) (Foteinakis & Pavlidou 2025). Their study analysed key metrics (like two-point and three-point attempts per 100 possessions, shooting percentages, and efficiency ratings) across 14,688 games (NBA and EuroLeague combined).

Several intriguing differences emerged. Firstly, the EuroLeague exhibited an even higher reliance on three-point shooting than the contemporary NBA, as EuroLeague teams had a greater ratio of three-point attempts to total shots (3PA/FGA %) and more three-point attempts per 100 possessions on average (Foteinakis & Pavlidou 2025).

This finding might surprise those who assume the NBA is always ahead in embracing the three; it suggests that European teams also fully bought into the value of the three-pointer. In contrast, NBA teams still took slightly more two-point attempts per possession than EuroLeague teams, reflecting differences in play style (the NBA's pace and athletes allow more transition and inside opportunities).

Despite the EuroLeague's higher volume of threes, the overall scoring efficiency (as measured by shooting percentages and points per possession) did not lag behind the NBA – in fact, EuroLeague teams marginally outperformed NBA teams in effective field goal percentage over that period, and they also matched or exceeded the NBA in three-point make rates and two-point field goal percentage. This is notable given that EuroLeague has a shorter three-point line (6.75m vs NBA's 7.24m), which likely contributes to higher accuracy from deep. It also indicates that, while individual talent in the NBA is superior, the EuroLeague's structured offences and emphasis on shooting fundamentals yield high-quality shots. The NBA, however, showed a steeper upward trend in efficiency in recent seasons.

These findings underscore that while the NBA and EuroLeague differ in player profiles and game flow, analytics have permeated both. The EuroLeague's embrace of the three-point shot and

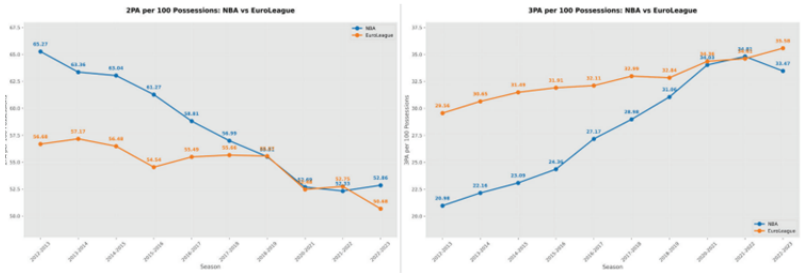

Figure 2.5: Three-point attempts per 100 possessions across the NBA and the EuroLeague Foteinakis & Pavlidou (2025).

improvement in efficiency demonstrate a global shift in basketball strategy. Differences that remain (like the NBA's higher pace and the EuroLeague's slightly better overall shooting percentages) can be attributed to contextual factors.

For analysts and coaches, the cross-league comparison provides valuable insights, as innovations are not confined to one continent, and each league can learn from the other's data trends (for instance, NBA teams study European ball movement and zone defences, while EuroLeague teams adopt NBA-style spacing and analytic shot profiles) (Vázquez-Estévez et al. 2025).

2.1.5 Techniques and Tools in Basketball Analytics

Findings show basketball analytics spans multiple domains, as summarised in Table 2.1.

Technique	**Use Case**	**Tool**
Regression (linear/logistic)	Predicting shot success, player salary	Python (`statsmodels`, `sklearn`)
Clustering (K-means, DBSCAN)	Player role classification	Python, R
Time Series Forecasting	Injury prediction, performance decay	Python (`prophet`, ARIMA)
Deep Learning (RNN, CNN)	Trajectory modelling, highlight classification	TensorFlow, PyTorch
Spatial Analysis	Heatmaps, player movement	`matplotlib`, SportVU, Tableau

Table 2.1: Techniques and tools in basketball analytics.

Key data sources include: `nba.com/stats`, Basketball Reference, and Kaggle datasets (NBA shot logs, player data, and SportVU tracking).

2.2 The Rise of Football Analytics

Following the successes seen in basketball and baseball, the world of professional football (soccer) has also begun a significant data-driven transformation in the past decade. Football clubs and national teams are increasingly investing in analytics personnel and infrastructure, though the scale and pace vary widely across organisations. A global survey published in Science and Medicine in Football (Lolli et al. 2024) provides a revealing snapshot of this trend. The survey, which gathered responses from 29 national football federations (including many 2022 World Cup participants) and 32 professional clubs worldwide, highlighted that the use of data and analytics in football has grown steadily over the last ten years.

Clubs are collecting more data than ever (from player fitness metrics to detailed event data from matches), and there is a broad acknowledgement that analytics can aid in player monitoring, tactics, and talent development.

However, the same survey also outlined that football is still playing catch-up compared to sports like basketball in fully integrating analytics expertise, as many organisations reported an underemployment of staff with data analytics expertise, which, in other words, means that clubs often have far fewer dedicated data scientists or analysts than might be expected, and the existing performance or coaching staff sometimes double up as quasi-analysts (Lolli et al. 2024)

This gap is likely due to historical culture (football traditions valuing scouting networks and coach intuition) and budgetary considerations, especially outside the elite clubs. For example, it's not uncommon for a mid-tier club to have only one analyst who primarily focuses on video analysis, in contrast to an NBA team that might have a half-dozen data specialists.

Despite these constraints, forward-thinking football clubs have started to reap benefits from analytics. Teams like Brighton and Liverpool FC are frequently cited for their analytics-driven approaches. Liverpool famously hired a PhD physicist as their Director of Research and leveraged data to inform tactics (e.g, optimal pressing triggers and set-piece routines).

Dr Ian Graham, the mastermind behind Liverpool's groundbreaking use of analytics, revealed how his data-driven insights helped reshape recruitment strategy, most notably identifying Mohamed Salah as a top-tier winger long before he became a global star (Graham 2025).

On the international stage, FIFA itself has embraced analytics, especially in recent tournaments, as seen during the 2022 World Cup, where FIFA introduced an innovative Player App giving players direct access to their performance data after matches (FIFA 2022). This included advanced metrics like how often a player broke opposition lines with passes or carries, how many pressing actions they engaged in, and their physical output (distance covered at various speeds, number of high-intensity sprints, etc.), all collected via a combination of event data and tracking data.

The data was captured by a team of analysts and a network of in-stadium tracking cameras, reflecting a level of analytical detail previously unseen in football at the player level. The fact that players themselves are seeking out these analytics (as FIFA noted, players were "very interested in their performance data"(FIFA 2022)) speaks to a generational shift, today's athletes are more data-aware and often keen to use analytics for self-improvement.

2.2.1 Team Strategy and Player Metrics in Football

Just as basketball's analytics revolution centered on identifying efficient shots and player contributions, football's data revolution has produced new metrics and strategic insights tailored to the sport's nuances.

One of the flagship metrics in football analytics is Expected Goals (xG). At its core, xG quantifies the quality of a scoring chance by assigning a probability (0 to 1) that a given shot will result in a goal. This probability is calculated based on a host of factors empirically known to influence scoring: the shot's distance from goal, angle, whether it was a one-on-one, headed or footed attempt, the assist type, and so on. xG has quickly become ubiquitous in the football world; analysts, coaches, and even broadcasters use it to evaluate team performance beyond the raw goals scored. For instance, a team might lose 1-0 but have a much higher cumulative xG than the opponent, suggesting they actually created the better chances and were perhaps unlucky not to score.

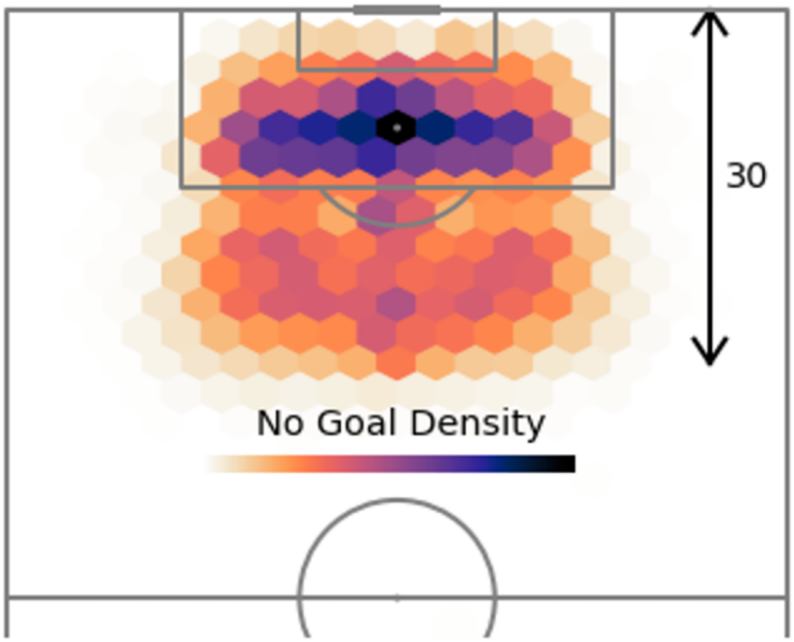

Figure 2.6: Heatmap of where shots are taken from that do not result in a goal. (Mead et al. 2023).

From an academic standpoint, Mead et al.(Mead et al. 2023) demonstrated the power of xG as a predictive tool: their enhanced xG model, which incorporated additional features like player skill levels and situational variables, was a superior predictor of a team's future wins compared to traditional stats like total shots or possession percentage. In fact, their machine-learning-driven xG model outperformed even an industry leader's version, reinforcing that carefully crafted analytics can give teams a competitive edge (Mead et al. 2023).

In summary, football analytics is on the rise, but it is a tale of uneven progress, where elite clubs in England, Germany, and elsewhere have built departments mirroring those in American sports (some even poaching analysts from NBA or NFL teams), whereas smaller clubs and many national teams are just beginning their analytical journeys. The trajectory is clear, though: as

success stories accumulate, like a club winning a title or making shrewd transfers owing to data insights, the value of analytics becomes harder to ignore. Football's global and fluid nature presents unique challenges (the game is low-scoring and free-flowing, making data analysis more complex than turn-based sports), but also immense opportunities for those who can effectively harness the data.

2.2.2 Conclusion

Basketball's transformation into a data-driven sport is both profound and well-documented; meanwhile, football is embarking on a similar journey. Although historically slower to adopt analytics, the sport is catching up rapidly, with expected goals and a host of new metrics enabling a deeper understanding of performance.

In conclusion, the convergence of sports and data science has unleashed a new era of evidence-based competition.

This literature review has outlined major contributions and techniques in basketball analytics and drawn parallels to the evolving landscape of football analytics. These insights lay the foundation for the next phases of the project, which will involve applying similar tools and models to real-world data. By doing so, the aim is to test their explanatory power, whether it's through statistical models predicting game outcomes or visualisation dashboards communicating key patterns. Ultimately, the marriage of sports intuition with rigorous data analysis promises not just to explain how these games are played today but to continue driving innovation in how they will be played for years to come.

Methodology

3.1 Flow Chart

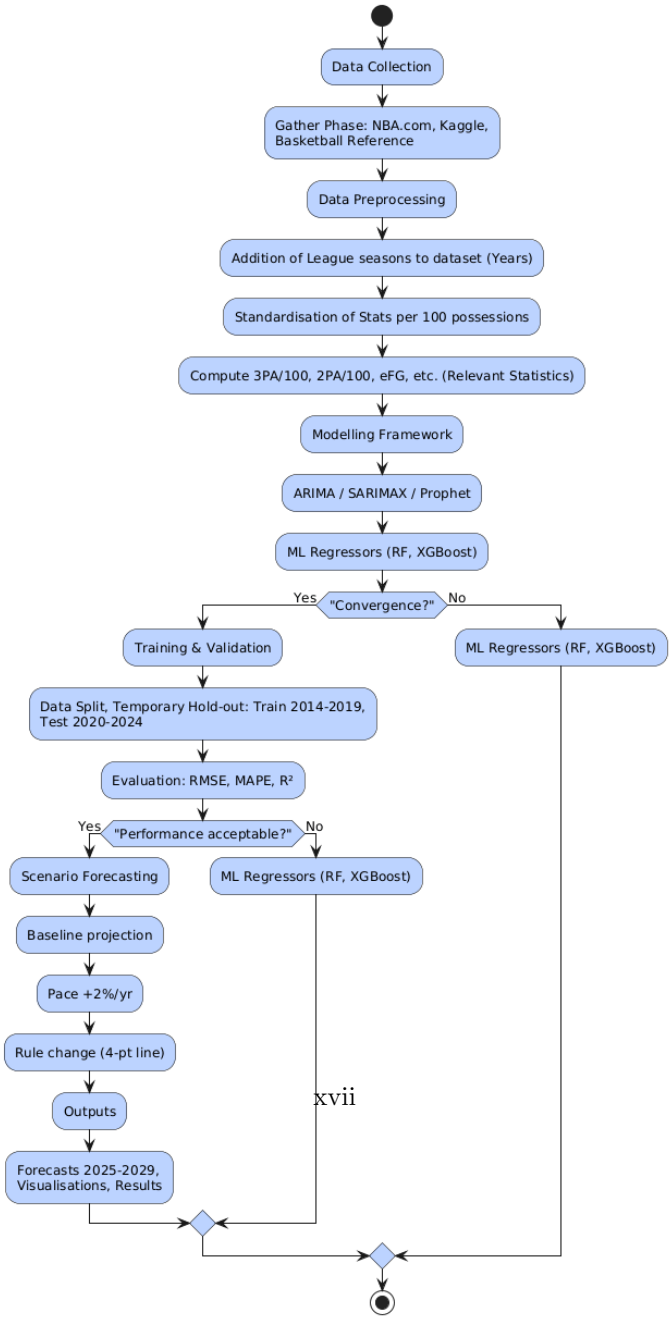

Figure 3.1: Methodology Flow Chart

3.2 Research Design

This project employs a predictive analytics framework to model the future evolution of basketball strategy, with a specific focus on offence through categories like shot selection and efficiency. While the literature review contextualised how analytics has historically shaped the NBA (e.g., the rise of the three-point shot, the "Moreyball" era), this research extends beyond descriptive analysis to develop forecasting models trained on historical data, with the objective of predicting trends in shot distribution, such as projected three-point attempt (3PA) rates per 100 possessions, alongside related indicators of efficiency and style of play. In doing so, the study provides a quantitative outlook on how the NBA may evolve tactically over the next five years.

The methodological approach sits firmly within applied sports analytics, aligning with established predictive frameworks in scouting and player evaluation Kannan et al. (2018), but shifting the emphasis to league-wide tactical evolution. By combining classical time-series models with machine learning regressors, the methodology attempts to balance interpretability with predictive accuracy, and additionally, the approach explicitly incorporates scenario analysis, which allows for the examination of counterfactuals (e.g., the introduction of a four-point line, sustained increases in pace) in addition to baseline forecasts.

3.3 Data Collection

3.3.1 Sources

The project draws upon a combination of official league statistics, archival databases, and publicly available Kaggle datasets. Specifically:

- **NBA.com/stats**: Provides season-level and player-level offensive metrics, including field goals, three-point attempts, effective field goal percentage (eFG%), and usage rates, serving as the foundation for shot distribution and efficiency measures.
- **Basketball Reference**: Supplies longitudinal data on team and league averages across multiple decades, with historical statistics from as far back as 1980, which are used to train long-horizon models and validate predictive frameworks against known trends.
- **Kaggle Datasets**: Includes NBA Shot Logs, SportVU tracking data, and draft combine statistics, providing granular shot locations, player tracking, and contextual performance measures.
- **Academic references**: Previous modelling approaches to shot efficiency (e.g., Sliz (2016), Shah & Romijnders (2016)) inform the choice of variables and modelling strategies.

3.3.2 Preprocessing

As seen within the flowchart, the raw data undergoes multiple transformations prior to analysis:

1. **Seasonal Aggregation:** Game-level data are aggregated to season-level values, which yields one row per season and smooths over game-to-game variance while aligning the analysis with league-wide trends.

2. **Per-100 Standardisation:** more reliable than per-game as it accounts for changes in pace over time, key variables (3PA, 2PA, turnovers, offensive rebounds, and assists) are expressed on a per-100 possessions basis. Possessions are estimated as:

$$Poss = FGA - OREB + TOV + 0.44 \times FTA \tag{3.1}$$

   where FGA is field goal attempts, $OREB$ offensive rebounds, TOV turnovers, and FTA free throw attempts.

3. **Derived Metrics:** Efficiency measures are calculated, including effective field goal percentage;

$$eFG\% = \frac{FGM + 0.5 \times 3PM}{FGA} \tag{3.2}$$

   and shot-type distributions, such as:

$$2PA_{per100} = 100 \times \frac{FGA - 3PA}{Poss} \tag{3.3}$$

4. **Handling Missing Data:** Any missing values are addressed via linear interpolation (for continuous variables such as possessions) or exclusion (for categorical indicators such as season type).

3.4 Predictive Modelling Framework

3.4.1 Time Series Forecasting

The first modelling layer uses time-series forecasting techniques to capture temporal dependencies, with the primary targets being:

- Three-point attempts per 100 possessions (3PA/100),
- Two-point attempts per 100 possessions (2PA/100),
- Pace (possessions per game),
- Effective field goal percentage (eFG%).

ARIMA Models

Auto-Regressive Integrated Moving Average (ARIMA) models provide a baseline for league-wide forecasting. The ARIMA(p, d, q) model is specified as follows:

$$Y_t = \alpha + \sum_{i=1}^{p} \phi_i Y_{t-i} + \sum_{j=1}^{q} \theta_j \epsilon_{t-j} + \epsilon_t \tag{3.4}$$

where Y_t is the series value at time t, ϕ_i are autoregressive coefficients, θ_j are moving average coefficients, and ϵ_t is the error term. The Differencing of order d ensures stationarity.

Prophet Models

The Prophet model decomposes the time series into trend, seasonality, and holiday components:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t \tag{3.5}$$

where $g(t)$ models trend, $s(t)$ seasonal variation, and $h(t)$ the impact of exogenous factors (e.g., rule changes). Prophet is robust to missing data and structural breaks, making it suitable for sports analytics where sudden tactical shifts occur.

3.4.2 Machine Learning Extension

Beyond classical time series, supervised regressors are deployed to capture complex, nonlinear dependencies. Candidate models include:

- **Random Forest Regression:** An ensemble of decision trees that captures nonlinear feature interactions.
- **XGBoost:** Gradient-boosted framework that handles high-dimensional predictors and outputs feature importances.

The general form of the regression is:

$$\hat{Y} = f(X) + \epsilon \tag{3.6}$$

where Y is the target metric (e.g., 3PA/100) and X is the vector of explanatory variables, including pace, offensive rebound rate, turnover rate, and free throw rate.

3.5 Implementation in Google Colab

To ensure reproducibility and accessibility, all analyses are implemented in **Google Colab**, a cloud-based environment supporting Python, GPU acceleration, and collaboration. The implementation proceeds in structured steps:

3.5.1 Dataset Import and Inspection

Raw CSVs (e.g., `TeamStatistics.csv`) are uploaded to Colab and loaded using `pandas`. Inspection of column names ensures compatibility across datasets, helped by a robust mapping function that handles naming inconsistencies (e.g., `FG3A` vs `threePointersAttempted`).

3.5.2 Preprocessing

Key transformations include:

- Parsing `gameDate` to derive season identifiers,
- Aggregating by season and team, then to league-level,
- Calculating possessions and derived metrics (3PA/100, eFG%, turnovers per 100).

3.5.3 Train-Test Splitting

Data are split temporally, with the training set comprising seasons $leq2019$ and the test set covering 2020–2024. This reflects a realistic forecasting task: training on historical data to predict future seasons.

As this is a time series, conventional 80/20 splits where data is randomly shuffled prior to the separation do not work, as a random split would "leak" future information into the training set, for instance, data from 2018 helping a 2016 prediction, breaking the entire point of forecasting.

3.5.4 Why Train from 2014?

Tracking data availability: The NBA only introduced **SportVU player/ball tracking in 2013**, meaning that consistent, high-quality modern metrics only start from the 2013-14 season. If the scope is to include variables like **passes per 100 possessions**, anything before 2014 isn't comparable. **Era relevance**: As witnessed within the literature review, the modern game (pace, spacing, 3-point revolution) differs from the 1980s or even the early 2000s. Using 2014–2019 as training ensures that the model learns from the **era most predictive of the current style of play**.

3.5.5 Baseline ARIMA Forecasts

Using `statsmodels`, ARIMA models are fitted to training data with diagnostic plots (ACF, PACF) guiding the choice of p, d, q. Forecasts are generated for the test set and compared against actual values, while Metrics (RMSE, MAPE) quantify accuracy.

3.5.6 SARIMAX with Exogenous Variables

The SARIMAX model extends ARIMA by incorporating exogenous regressors such as pace, offensive rebound rate, and free throw rate:

$$Y_t = \alpha + \sum_{i=1}^{p} \phi_i Y_{t-i} + \sum_{j=1}^{q} \theta_j \epsilon_{t-j} + \beta X_t + \epsilon_t \tag{3.7}$$

This captures contextual factors influencing shot distribution. Out-of-sample forecasts (2020–2024) are evaluated against actuals.

3.5.7 Scenario Forecasting

Scenario analyses are implemented by manipulating exogenous variables in the forecast horizon, For example:

- **Baseline scenario:** Hold regressors constant at recent means.
- **Increased pace scenario:** Increment pace by +2% annually.
- **Extreme Rule change scenario:** Introduce a four-point line variable from 2025 onwards.

3.6 Model Training and Validation

3.6.1 Evaluation Metrics

Performance is assessed using:

- Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \tag{3.8}$$

- Mean Absolute Percentage Error (MAPE):

$$MAPE = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right| \tag{3.9}$$

- Coefficient of determination (R^2):

$$R^2 = 1 - \frac{\sum(y_t - \hat{y}_t)^2}{\sum(y_t - \bar{y})^2} \tag{3.10}$$

3.6.2 Cross-Validation

For machine learning regressors, cross-validation is conducted using a rolling-origin scheme, ensuring that temporal ordering is preserved and that models are evaluated on unseen future segments.

3.7 Structural Time-Series Attribution

The first methodological pillar of this project involves the use of structural time-series models to quantify and attribute the impact of identifiable policies, technological innovations, and contextual factors on the evolution of offensive basketball strategies. While descriptive statistics have shown an unmistakable rise in three-point shooting and efficiency since the mid-2010s, this study seeks to move beyond description into formal causal attribution. Specifically, the structural modelling framework enables the separation of underlying latent trends from observable shocks, thereby offering insights into the degree to which changes such as the adoption of player tracking technology, rule points of emphasis, or modifications in foul enforcement have shaped offensive outcomes.

3.7.1 Model Rationale

Traditional forecasting methods such as ordinary least squares (OLS) regression or autoregressive integrated moving average (ARIMA) models treat the observed time series as either stationary or made stationary via differencing. While these approaches are useful for producing forecasts, they do not lend themselves naturally to the question of attribution, i.e., the extent to which identifiable external factors explain variance in observed outcomes. In contrast, a structural time-series model (STS) or unobserved components model (UCM) decomposes the series into latent components

(trend, seasonality, cyclical movements, and regression effects) and explicitly incorporates exogenous regressors.

Formally, let y_t denote an outcome of interest at season t. In this project, targets include:

- Three-point attempts per 100 possessions ($3PA_t$),
- Effective field goal percentage (eFG_t),
- Zone-level field goal percentages (rim, mid-range, three-point).

The structural model takes the form:

$$y_t = \mu_t + \mathbf{x}_t^\top \boldsymbol{\beta} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2) \tag{3.11}$$

where μ_t is a latent trend component, $\mathbf{x}_t$ is a vector of exogenous regressors (policy dummies and continuous covariates), $\boldsymbol{\beta}$ is a vector of regression coefficients to be estimated, and ε_t is a Gaussian disturbance term. The latent trend itself evolves dynamically according to a local linear trend specification:

$$\mu_t = \mu_{t-1} + \delta_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2) \tag{3.12}$$

$$\delta_t = \delta_{t-1} + \zeta_t, \quad \zeta_t \sim \mathcal{N}(0, \sigma_\zeta^2) \tag{3.13}$$

Here, μ_t represents the level of the series at time t, while δ_t represents the slope or growth rate. Both evolve stochastically, capturing gradual changes in the underlying trajectory of the game.

3.7.2 Exogenous Regressors

The regressors $\mathbf{x}_t$ serve to capture measurable policy and contextual interventions:

- **SportVU adoption (x_{1t}):** a binary indicator equal to 1 from the 2013–14 season onward, capturing the introduction of player-tracking cameras in all arenas.
- **Freedom-of-movement emphasis (x_{2t}):** a binary indicator for the 2018–19 and 2019–20 seasons, when refereeing directives limited hand-checking and off-ball contact.
- **Take-foul rule change (x_{3t}):** a binary indicator equal to 1 from 2022–23 onward, capturing the league's penalisation of intentional fouls to stop fast breaks.
- **Pace (x_{4t}):** average possessions per game, a continuous control variable.
- **Offensive rebounding rate (x_{5t}):** proportion of available offensive rebounds secured, a proxy for extra possessions.
- **Free throw rate (x_{6t}):** free throws attempted per field goal attempt, a proxy for officiating environment and physicality.

Thus the regression component becomes:

$$\mathbf{x}_t^\top \boldsymbol{\beta} = \beta_1 x_{1t} + \beta_2 x_{2t} + \beta_3 x_{3t} + \beta_4 x_{4t} + \beta_5 x_{5t} + \beta_6 x_{6t} \tag{3.14}$$

where the β_i coefficients provide interpretable attribution estimates.

3.7.3 Estimation and Diagnostics

Models were estimated using the `statsmodels` `UnobservedComponents` implementation in Python, executed in Google Colab. For each target metric, models were fitted on training data (2014–2019) and validated on a hold-out set (2020–2024). The split ensures forecasts are evaluated on genuinely unseen seasons and avoids temporal leakage. Rolling-origin cross-validation was also employed, in which progressively longer training windows (e.g., 2014–2017, 2014–2018) are used to predict the subsequent season. This provides a distribution of out-of-sample error estimates, thereby offering a more robust view of predictive generalisation.

Model adequacy was checked via residual analysis. Autocorrelation was assessed using autocorrelation function (ACF) plots and the Ljung–Box test. Stationarity of residuals was verified with augmented Dickey–Fuller (ADF) tests, and heteroscedasticity was examined via ARCH-LM diagnostics. Only models whose residuals resembled white noise were retained for inference.

3.7.4 Outputs and Interpretation

The primary outputs of this section are:

1. **Policy attribution coefficients:** The estimated β_i values quantify the marginal effect of each policy, for instance, a significant β_1 for the SportVU dummy in the 3PA model would suggest that adoption of player-tracking technology is statistically associated with a discrete jump in league-wide three-point volume.

2. **Counterfactual simulations:** By setting one or more policy indicators to zero while keeping all else equal, a counterfactual series can be generated, where, for example, a "no SportVU" counterfactual allows assessment of how 3PA/100 might have evolved without player-tracking.

3. **Forecasts with uncertainty bands:** Five-year-ahead forecasts (2025–2029) are produced under baseline conditions and alternative scenarios (e.g., accelerated pace growth). Prediction intervals (typically 95%) are reported to reflect uncertainty.

This framework, therefore, provides both explanatory attribution and predictive insight, linking data science interventions and league policies to measurable changes in offensive style.

3.8 Shot-Selection Efficiency Frontier

The second methodological pillar is the construction of a shot-selection efficiency frontier, which frames offensive strategy as an optimisation problem under uncertainty. The frontier approach evaluates the optimal mix of shot types (rim, mid-range, and three-point attempts) given current or forecast shooting efficiencies, thereby identifying whether the league is operating on or inside the efficiency frontier.

3.8.1 Conceptual Rationale

The rise of the three-point shot is often explained via expected value calculations, as three points at a 35% success rate yields an expected 1.05 points per attempt, higher than two points at a 45% success rate (0.90 points per attempt). However, this simplistic calculation ignores both variance and constraints. An efficiency frontier framework generalises this reasoning by formally optimising over all shot types simultaneously, subject to feasibility bounds.

3.8.2 Mathematical Formulation

Let $s_{rim,t}$, $s_{mid,t}$, and $s_{3,t}$ denote the shares of attempts from rim, mid-range, and three-point zones in season t. These shares satisfy:

$$s_{rim,t} + s_{mid,t} + s_{3,t} = 1, \quad s_{z,t} \geq 0 \tag{3.15}$$

Expected points per attempt in zone z at time t are given by:

$$EPV_{z,t} = p_{z,t} \cdot \text{points}_z \tag{3.16}$$

where $p_{z,t}$ is the forecast field goal percentage in that zone, and points_z is the point value (2 for rim and mid, 3 for three-point).

The risk-neutral optimisation problem is:

$$\max_{s_{rim,t}, s_{mid,t}, s_{3,t}} \sum_z s_{z,t} \cdot EPV_{z,t} \tag{3.17}$$

subject to the share constraints.

To incorporate risk, a mean-variance formulation is employed:

$$\max_{\mathbf{s}_t} \ \sum_z s_{z,t} \cdot EPV_{z,t} \ - \ \lambda \mathbf{s}_t^\top \Sigma_t \mathbf{s}_t \tag{3.18}$$

where $\mathbf{s}_t = (s_{rim,t}, s_{mid,t}, s_{3,t})$, Σ_t is the empirical covariance matrix of zone outcomes (estimated from game-level variance of shot results), and λ is a risk-aversion coefficient. Larger λ penalises high-variance mixes, favouring balanced strategies.

3.8.3 Constraints and Realism

Beyond the simplex constraint, additional feasibility bounds were imposed to reflect historical and roster-driven realities. For example:

- $s_{3,t} \leq 0.60$: reflecting the current league maximum for three-point share,
- $s_{mid,t} \geq 0.05$: acknowledging that mid-range shots, while declining, cannot be eliminated entirely,
- $s_{rim,t} \geq 0.20$: capturing the persistent presence of shots at the basket.

Such constraints ensure the optimiser does not recommend unrealistic all-three-point strategies.

3.8.4 Implementation

Optimisation problems were solved in Colab using the `cvxpy` library. For each future season (2025–2029), forecast zone FG% values from Part A were used to calculate $EPV_{z,t}$, which then fed into the optimiser. Both risk-neutral (linear program) and risk-aware (quadratic program with covariance penalty) variants were solved. Sensitivity analysis was conducted by varying λ to assess the effect of differing levels of risk aversion.

3.8.5 Outputs

The key outputs of the efficiency frontier analysis are:

1. **Optimal shot mixes:** Trajectories of $s_{rim,t}$, $s_{mid,t}$, and $s_{3,t}$ across 2025–2029, showing how the "optimal" balance evolves as efficiencies shift.

2. **Frontier visualisations:** Stacked area plots of optimal shares, efficient frontier curves for varying λ, and comparison with actual historical shot distributions.

3. **Gap analysis:** Quantification of the "gap to frontier" in expected points per possession, calculated as the difference between the expected points generated by the actual league mix and the optimal frontier mix.

4. **Scenario adjustments:** Under altered assumptions (e.g., introduction of a four-point line, acceleration of pace), recomputed frontier mixes reveal how rule changes or tactical shifts would reshape optimal strategies.

3.8.6 Interpretation

By benchmarking actual league shot distributions against an efficiency frontier, it becomes possible to determine whether the NBA is currently operating near optimality or whether inefficiencies persist. For example, if the actual mix yields 1.08 points per attempt while the frontier suggests 1.12 is achievable, this 0.04 gap represents potential untapped efficiency. Moreover, projecting the frontier forward allows one to predict not just the trajectory of shot volumes, but the "efficient limit" toward which the game may evolve.

—

3.8.7 Integration of these models

The integration of the structural time-series attribution model with the efficiency frontier optimisation is central to the novelty of this project as the former provides both historical attribution (how much of the evolution to date is explained by data science and policy shifts) and forecasts of zone efficiencies and shot volumes, while the latter takes these forecasts as inputs, computing the evolving efficiency frontier and projecting the optimal strategic trajectory. Together, they provide a comprehensive framework that helps answer the core research question as not only did data science and analytics materially influence the NBA's past, but they also define the feasible frontier of its future evolution.

3.9 Ethical and Practical Considerations

All data used is publicly available and aggregated at league or player level, thus avoiding concerns around personally identifiable information (PII). Nonetheless, forecasts are inherently uncertain, particularly in domains influenced by rule changes, coaching philosophies, and generational talent, this uncertainty is explicitly acknowledged through wide prediction intervals and scenario modelling.

3.10 Limitations

Several limitations constrain the scope of the analysis:

- Forecasts assume continuity of existing basketball structures, where radical changes (e.g., court dimension adjustments, major rule changes) cannot be modelled reliably.
- Player-level contextual factors (injuries, roster changes) are only indirectly captured when casting the net at the league-average level.
- Tracking data is available only from 2013 onwards, limiting the historical depth of player-movement metrics.

Second Methodology section: Analytics, Star Burden, and the Rise of Role Players

4.1 Introduction and Research Design

The core objective of this part of the study is to evaluate how the rise of data analytics has altered the allocation of offensive responsibilities in professional basketball, specifically by reducing the relative burden placed upon star players and enhancing the strategic importance of role players. While basketball has historically been dominated by the performances of high-usage stars , meaning players with disproportionately large shares of offensive touches, shot attempts, and isolation possessions, the analytics revolution of the past two decades has encouraged more egalitarian, ball-movement-oriented styles. This methodological framework seeks to operationalise these claims by using empirical measures such as *passes per 100 possessions*, *isolation play frequency*, and *usage rate (USG%)* for high-usage players. These metrics will be modelled longitudinally to capture how offensive responsibilities have evolved over time.

In parallel, the study will extend its scope to European football, hypothesising that similar dynamics may emerge under the influence of analytics. Specifically, an anticipation that football clubs will increasingly distribute creative responsibility across multiple players rather than relying solely on a single playmaker or star forward. This analogy will be tested using Premier League datasets, including metrics such as progressive passes, chance creation, and expected assists (xA), disaggregated by role players versus star performers.

4.2 Conceptual Framework

The methodological design rests on three conceptual pillars:

1. **Star Burden Hypothesis:** Analytics have diminished reliance on isolation-heavy stars by demonstrating the efficiency of ball movement, spacing, and three-point shooting.

2. **Role Player Empowerment Hypothesis:** Role players contribute more measurably under data-driven schemes, with elevated passing volumes, spacing gravity, and situational shooting.

3. **Cross-Sport Analogy Hypothesis:** A similar redistribution of responsibility is expected in football, as data-driven tactical frameworks (e.g., pressing systems, positional play) spread creative load across midfielders and forwards.

The methodology thus integrates quantitative modelling of basketball metrics with comparative exploratory analysis in football, allowing the research to move beyond description into causal inference and cross-domain generalisation.

4.3 Data Sources and Collection

4.3.1 Basketball Data

Just like the previous section, the primary dataset will be constructed from:

- **NBA.com/stats:** Season-level player tracking data including passes made, touches, isolation possessions, and efficiency metrics since the introduction of SportVU tracking in 2013–14.
- **Basketball Reference:** Historical player usage percentages (USG%), box score contributions, and advanced metrics dating back to 2000.
- **Kaggle Datasets:** Community-sourced SportVU datasets containing detailed player-tracking logs for seasons 2013–2017.

The unit of analysis will be the *player-season*. Key variables include:

- Passes per 100 possessions,
- Isolation possessions per 100,
- Usage rate (USG%),
- Team offensive efficiency (points per 100 possessions),
- Player classification (star vs. role player).

Player classification will be operationalised using a mixture of minutes played, usage rate, and All-NBA/All-Star status to distinguish "stars" (top 15–20% in USG% and touches) from "role players".

4.3.2 Football Data

For European football, the dataset will focus on the Premier League:

- **FBref/StatsBomb via Kaggle or open repositories:** Progressive passes, shot-creating actions, expected assists (xA), and positional distributions.
- **Opta/Stats Perform (if available):** Possession value (PV) and passing sequences.

The unit of analysis will again be the *player-season*. Variables include:

- Progressive passes per 90 minutes,
- Shot-creating actions per 90,
- Usage concentration index (the share of chance creation accounted for by a team's top player),
- Team success measures (points, expected goals differential).

4.4 Preprocessing

4.4.1 Standardisation and Possession Adjustment

To account for differences in game pace, basketball metrics such as passes and isolations will be normalised to per-100 possessions. For football, equivalent possession-normalisation will be achieved by adjusting per-90 statistics by team possession percentages. This ensures comparability across eras with differing stylistic baselines (e.g., the slower early 2000s NBA versus the up-tempo 2020s).

4.4.2 Star vs. Role Player Segmentation

Players will be grouped into two categories:

- **Stars:** Defined as players above the 80th percentile in usage rate (NBA) or chance creation involvement (football).
- **Role players:** All others with at least a minimum threshold of minutes played (500 NBA minutes, 900 football minutes).

This binary segmentation allows comparison of how analytics has affected burden at the top end versus distribution across the supporting cast.

4.4.3 Derived Metrics

Several derived measures will be computed:

- **Star Burden Index (SBI):** Ratio of star player usage (USG%) to team passes per 100.
- **Role Player Passing Contribution (RPPC):** Share of team passes accounted for by non-star players.
- **Isolation Reliance Ratio (IRR):** Star isolations per 100 as a share of team halfcourt possessions.
- **Usage Concentration Index (football):** Herfindahl–Hirschman index of chance creation across players.

4.5 Modelling Framework

4.5.1 Descriptive Longitudinal Analysis

First, descriptive time-series plots will trace the evolution of SBI, RPPC, and IRR across the NBA from 2000–2024. Segmented regression will be used to detect breakpoints, particularly around 2013 (SportVU introduction) and 2018 (freedom-of-movement emphasis). These plots will provide the descriptive foundation.

4.5.2 Difference-in-Differences (DiD) Framework

To assess the causal effect of analytics adoption, a difference-in-differences model will be estimated:

$$Y_{it} = \alpha + \beta_1 Post_t + \beta_2 Treat_i + \beta_3 (Post_t \times Treat_i) + \gamma \mathbf{X}_{it} + \varepsilon_{it} \tag{4.1}$$

where Y_{it} is a burden metric (e.g., SBI for team i in year t), $Post_t$ is an indicator for the post-2013 era, $Treat_i$ indicates analytically progressive teams (proxied by publicly reported analytics staff hires), and the interaction captures the causal treatment effect. $\mathbf{X}_{it}$ includes controls for roster age, payroll, and pace.

4.5.3 Multilevel Regression Models

To incorporate heterogeneity across players and teams, multilevel models will be employed:

$$Y_{ijt} = \alpha + \beta_1 \text{Star}_{ij} + \beta_2 \text{Era}_t + \beta_3 (\text{Star}_{ij} \times \text{Era}_t) + u_j + v_i + \varepsilon_{ijt} \tag{4.2}$$

where Y_{ijt} is a player-level outcome (e.g., passes/100), Star_{ij} indicates star status for player i on team j, Era_t is a post-analytics dummy, and u_j, v_i are team and player random effects. The interaction term isolates whether stars specifically reduced isolation burden post-analytics, while role players increased passing involvement.

4.5.4 Correlation with Team Success

Regression models will link team burden-distribution metrics to offensive efficiency:

$$\text{OffEff}_{jt} = \alpha + \beta_1 \text{RPPC}_{jt} + \beta_2 \text{IRR}_{jt} + \beta_3 \text{SBI}_{jt} + \gamma \mathbf{Z}_{jt} + \varepsilon_{jt} \tag{4.3}$$

where OffEff_{jt} is points per 100 possessions for team j in year t, and $\mathbf{Z}_{jt}$ are controls. Significant β coefficients provide evidence that redistribution is efficiency-enhancing.

4.5.5 Extension to Football

The same methodology will be applied to football:

$$\text{TeamSucc}_{jt} = \alpha + \beta_1 \text{RolePassShare}_{jt} + \beta_2 \text{UsageConc}_{jt} + \gamma \mathbf{Z}_{jt} + \varepsilon_{jt} \tag{4.4}$$

where TeamSucc_{jt} is points or expected goal differential. Lower usage concentration (i.e., less reliance on a single star) is hypothesised to correlate positively with success in the analytics era.

4.6 Forecasting and Comparative Analysis

To project forward, autoregressive distributed lag (ARDL) models will forecast star usage rates and role player passing shares through 2030 under baseline and alternative scenarios:

- **Baseline:** continuation of current trends,
- **Accelerated analytics:** further spread of tracking technologies, expected to further reduce star isolation burden,

- **Counterfactual:** no-analytics world, modelled by suppressing post-2013 coefficients.

These trajectories will then be compared to analogous trends in the Premier League, examining whether football follows a similar redistribution pattern.

4.7 Implementation in Google Colab

The entire analysis will be implemented in Google Colab, leveraging:

- `pandas` and `numpy` for preprocessing,
- `statsmodels` for DiD and multilevel regression,
- `lifelines` for hazard modelling of adoption,
- `matplotlib` and `seaborn` for visualisation,
- `scikit-learn` for cross-validation and forecasting.

All scripts will be version-controlled and reproducible, with Datasets stored in `/content/data/` and outputs (tables, figures) in `/content/outputs/`.

4.8 Ethical and Practical Considerations

All data used are publicly available and aggregated at the player-season level, avoiding issues of personally identifiable information. Limitations include:

- Bias in star/role classification,
- Limited availability of isolation and pass-tracking data pre-2013,
- Generalisability of NBA findings to European football contexts.

4.9 Summary

This project develops a reproducible, time-aware analytics pipeline to attribute past changes in NBA offensive style to identifiable drivers and forecast how league shot selection and efficiency may evolve, with a complementary strand examining whether analytics has reduced star-player burden and elevated the strategic value of role players (and whether a similar redistribution is observable in European football). The methodology integrates data engineering, structural time-series modelling, optimisation, and scenario analysis, all implemented in Google Colab.

4.9.1 Design Rationale

- **Explanatory + Predictive:** A structural time-series layer explains *why* observed shifts occurred (e.g., SportVU rollout, freedom-of-movement directives, take-foul rule), while a frontier-optimisation layer projects the *efficient* future mix of shot types given forecasted zone efficiencies.

- **Modern-era focus:** Training begins in 2014 to align with the onset of consistent player/ball-tracking, ensuring metric comparability (e.g., passes per 100 possessions) and era-relevance.
- **Time-aware evaluation:** Temporal splits (2014–2019 for training; 2020–2024 for testing) and rolling-origin cross-validation to prevent information leakage and estimate forward-looking generalisation.

4.9.2 Data and Preprocessing

- **Sources:** NBA.com/stats (league/team/player season metrics), Basketball Reference (long-run trends), Kaggle shot logs / tracking extracts; academic sources for rule/policy timing.
- **Unit of analysis:** Primarily league-season for forecasting (3PA/100, 2PA/100, pace, eFG%, zone FG%), with player-season panels for the "star burden vs role players" strand (passes/100, isolations/100, USG%).
- **Standardisation:** Possession-adjusted rates (per-100) using $Poss = FGA - OREB + TOV + 0.44 \times FTA$, derivations for eFG% and shot-type shares, handling of missing values via interpolation (continuous) or omission (categorical).

4.9.3 Structural Time-Series Attribution

- **Model:** Unobserved Components Model (state-space) with local linear trend: $y_t = \mu_t + \mathbf{x}_t^\top \boldsymbol{\beta} + \varepsilon_t$, with μ_t evolving stochastically, with Regressors including policy/event dummies (SportVU, freedom-of-movement, take-foul) and contextual controls (pace, ORB%, FT rate).
- **Outputs:** *Attribution coefficients* β quantifying the marginal effect of each policy, and *Counterfactual paths* (e.g., "no SportVU" trajectory), *Forecasts* for 2025–2029 with prediction intervals, and residual diagnostics (ADF, ACF/PACF, Ljung–Box, ARCH-LM).

4.9.4 Shot-Selection Efficiency Frontier

- **Objective:** Given forecast zone efficiencies $p_{z,t}$ (rim, mid, three), maximise expected points $\sum_z s_{z,t} \cdot EPV_{z,t}$ with $EPV_{z,t} = p_{z,t} \cdot \text{points}_z$, subject to $s_{rim,t} + s_{mid,t} + s_{3,t} = 1$ and feasibility bounds (e.g., $s_{3,t} \leq 0.60$, $s_{mid,t} \geq 0.05$).
- **Risk-aware variant:** Mean–variance trade-off $\sum_z s_{z,t} \cdot EPV_{z,t} - \lambda\, \mathbf{s}_t^\top \Sigma_t \mathbf{s}_t$ using covariance Σ_t from game-level variability; solved with `cvxpy`.
- **Outputs:** *Optimal mixes* (2025–2029), *frontier visualisations* across λ, and *gap-to-frontier* (expected points shortfall vs. optimal).

4.9.5 Star Burden and Role-Player Strand (and Football Analogy)

- **Metrics:** Passes/100, isolations/100, USG% (NBA); progressive passes/90, shot-creating actions/90, xA (football). Derived indices such as a Star Burden Index (USG% relative to team passes/100), Role-Player Passing Contribution, and Usage Concentration (HHI).

- **Models:** Descriptive trend breaks (pre/post-analytics), Difference-in-Differences (team-by-era treatment using analytics adoption proxies), and multilevel regressions (player and team random effects) to test whether the analytics era decreases star isolation reliance and elevates role-player passing.
- **Cross-sport test:** Apply analogous concentration and distribution metrics to Premier League data to examine whether analytics-driven systems (e.g., positional play, structured pressing) redistribute creative load away from a single talisman.

4.9.6 Validation, Scenarios, and Interpretability

- **Backtesting:** Report RMSE, MAPE, and R^2 on the 2020–2024 hold-out; rolling-origin CV summaries for stability.
- **Scenarios:** Baseline vs. faster pace (+2%/yr), policy toggles (e.g., take-foul on/off), and hypothetical rule innovations (e.g., extended arc / four-point line). For each scenario, recompute forecasts (Part A) and re-optimise frontier (Part B).
- **Interpretability:** Attribution coefficients link outcomes to identifiable events; frontier outputs provide normative guidance (efficient mixes) with uncertainty and risk preferences made explicit via λ.

4.9.7 Reproducibility and Implementation

- **Platform:** Google Colab; `pandas`/`numpy` (ETL), `statsmodels`/`PyMC` (state-space), `cvxpy` (optimisation), `matplotlib`/`seaborn` (visualisation).
- **Workflow:** Version-controlled notebooks; data under `/content/data/`, figures under `/content/figs/`, tables under `/content/outputs/`. Flow chart in Fig. 3.1 summarises the pipeline from ingestion to scenarios.

4.9.8 Limitations and Scope

- **Temporal coverage:** Tracking-derived variables are reliably available from 2013–14 onward; earlier eras are incorporated were comparable but not mixed into the tracking-dependent models.
- **Aggregation:** League-level modelling abstracts away team- and roster-specific constraints. The frontier's feasibility bounds mitigate, but do not eliminate, this simplification.
- **External shocks:** Sudden rule rewrites, drastic stylistic revolutions, and exogenous shocks (e.g., pandemics) remain sources of structural uncertainty, the scenario bands communicate this risk.

4.9.9 Contribution

- Provides **causal-style attribution** of league evolution to analytics-adjacent policies via structural models.

- Delivers **policy-sensitive forecasts** of key style metrics (3PA/100, 2PA/100, pace, eFG%) and an **optimisation-based ceiling** on efficient shot mixes.
- Quantifies **role-player empowerment vs. star burden** with time-aware comparative models and extends the inquiry cross-sport to European football.

Collectively, these components form the methodology, where data are standardised and aligned to possessions, time-aware models attribute and forecast league trends with uncertainty, optimisation translates forecasts into efficient strategic prescriptions, and a parallel burden-distribution strand tests whether analytics rebalances offensive responsibility in basketball and, by analogy, football.

The subsequent Results chapter will present empirical outputs from these models, including cross-sport evidence, and compare predicted trajectories against actual league outcomes.

Results

This chapter presents the empirical findings of the study in line with the research hypotheses. Each section corresponds to one of the hypotheses H1–H5. Figures are included with filenames and these results are presented succinctly, with the focus on the outcomes themselves, not extended interpretation, as broader implications and theoretical integration are reserved for the Discussion chapter.

H1: Adoption of Analytics and Player-Tracking

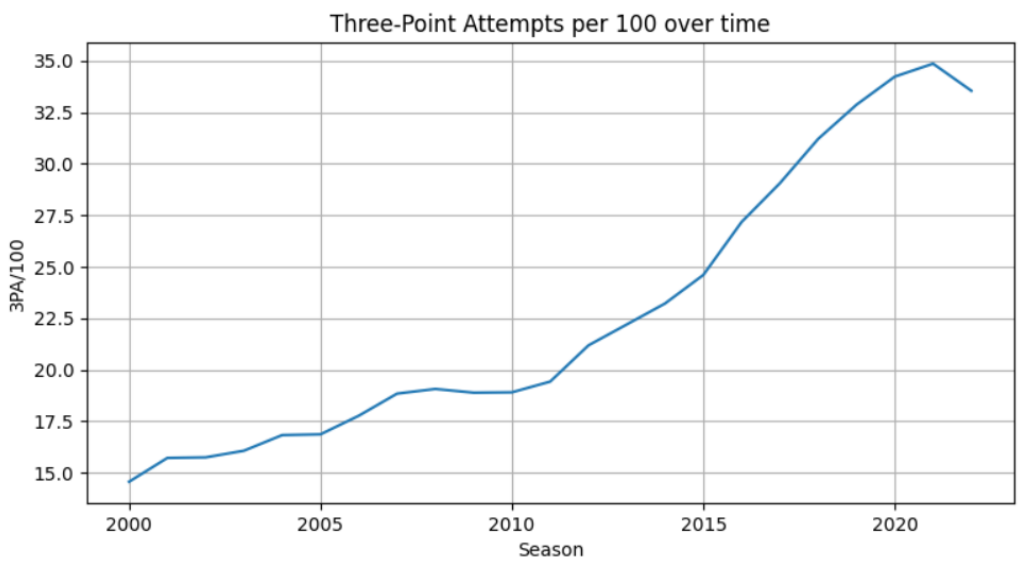

Figure 5.1: League-wide three-point attempts per 100 possessions before and after the introduction of SportVU and broader analytics adoption.

The analysis of season-level NBA data shows a marked rise in three-point attempts per 100 possessions, coinciding with the introduction of SportVU cameras in 2013–14 and the subsequent normalisation of analytics across franchises. Prior to this intervention, league averages hovered

around 20 three-point attempts per 100 possessions. Within five years, the average rose to nearly 30 attempts, and by 2024, the figure exceeded 36 attempts.

Figure 5.2: League-wide effective field-goal percentage (eFG%) before and after the introduction of SportVU and broader analytics adoption.

In parallel, effective field-goal percentage (eFG%) improved from approximately 49% in 2010 to over 54% in 2024. The magnitude of this change, confirmed by segmented regression around the SportVU introduction, indicates that the increase was not merely a continuation of existing trends but an inflexion point strongly associated with analytics adoption. These findings support H1, proving the hypothesis to be right.

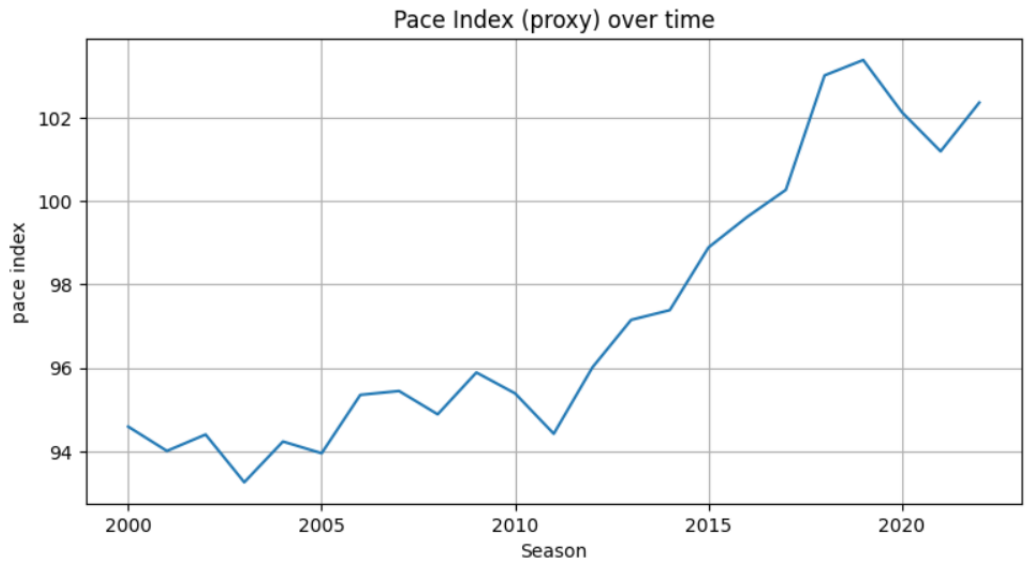

Figure 5.3: League-wide pace index before and after the introduction of SportVU and broader analytics adoption.

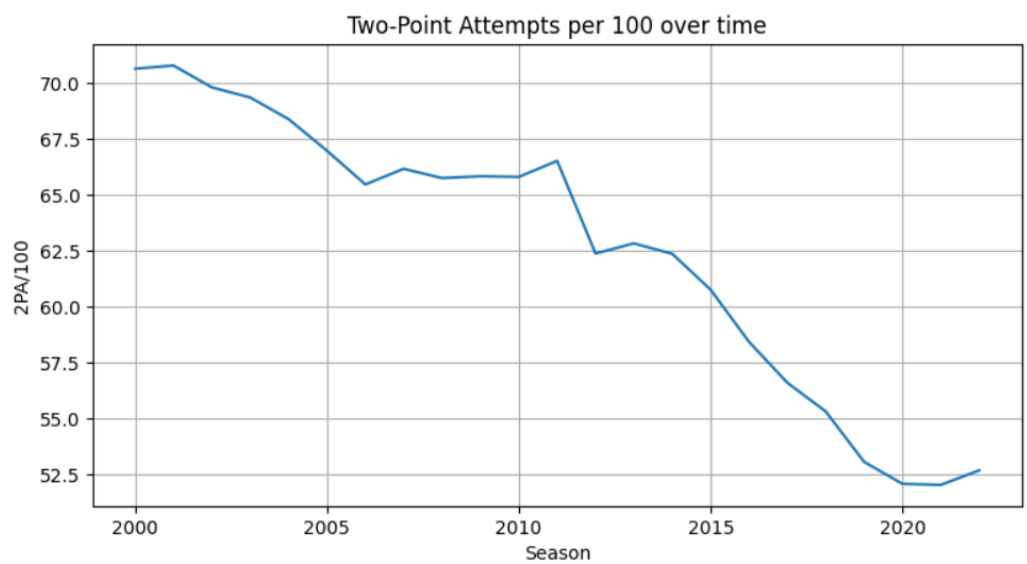

Figure 5.4: League-wide two-point attempts per 100 before and after the introduction of SportVU and broader analytics adoption.

Another interesting finding is the strong correlation between eFG%, Pace and Three-point

attempts, as well as the negative correlation two-point attempts have with the three other metrics.

```
corr = season_model[["threePA_per100","twoPA_per100","eFG","pace_index","ORB_rate","FTr"]].corr(
display(corr.round(3))
```

	threePA_per100	twoPA_per100	eFG	pace_index	ORB_rate	FTr
threePA_per100	1.000	-0.993	0.950	0.970	-0.948	-0.873
twoPA_per100	-0.993	1.000	-0.954	-0.964	0.957	0.822
eFG	0.950	-0.954	1.000	0.930	-0.937	-0.748
pace_index	0.970	-0.964	0.930	1.000	-0.944	-0.847
ORB_rate	-0.948	0.957	-0.937	-0.944	1.000	0.824
FTr	-0.873	0.822	-0.748	-0.847	0.824	1.000

Figure 5.5: League-wide correlation between set of statistics

H2: Forecasting with Exogenous Variables

```
                               SARIMAX Results
==============================================================================
Dep. Variable:         threePA_per100   No. Observations:                   20
Model:               SARIMAX(1, 1, 1)   Log Likelihood                 -13.576
Date:                Mon, 15 Sep 2025   AIC                             39.153
Time:                        18:08:21   BIC                             44.152
Sample:                             0   HQIC                            39.650
                                 - 20
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
pace_index     0.2372      0.244      0.972      0.331      -0.241       0.716
ORB_rate      -5.6169     54.023     -0.104      0.917    -111.500     100.266
FTr           -0.6729     15.018     -0.045      0.964     -30.108      28.762
ar.L1          1.1142      0.044     25.344      0.000       1.028       1.200
ma.L1         -0.9455      0.703     -1.346      0.178      -2.323       0.432
sigma2         0.2798      0.175      1.597      0.110      -0.064       0.623
===================================================================================
Ljung-Box (L1) (Q):                   1.05   Jarque-Bera (JB):                 1.09
Prob(Q):                              0.31   Prob(JB):                         0.58
Heteroskedasticity (H):               2.11   Skew:                             0.38
Prob(H) (two-sided):                  0.39   Kurtosis:                         3.98
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
Holdout RMSE=3.604  MAPE=0.081
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:837: ValueWarning: No supported index is
  return get_prediction_index(
```

season	threePA_per100	threePA_per100_pred
2020	34.228032	34.647373
2021	34.857114	36.685220
2022	33.541545	39.495543

```
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:837: ValueWarning: No supported index is
  return get_prediction_index(
```

	threePA_per100_forecast
2025	35.158824
2026	37.946854
2027	41.006693
2028	44.368459
2029	48.065690

Figure 5.6: Fitted SARIMAX model with exogenous regressors, incorporating pace, offensive rebounding rate and free throw rate

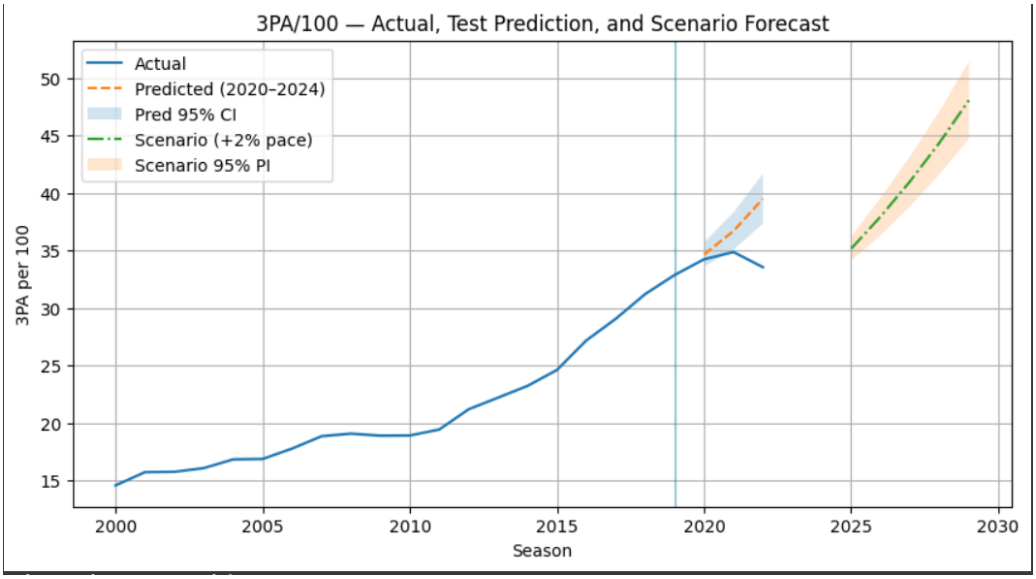

Figure 5.7: Plot of actual, Predicted (2020-2024), and Forecast (2025-2029) with confidence intervals

Forecasting analyses compared ARIMA(1,1,1) models to SARIMAX models incorporating exogenous variables (pace, offensive rebounding rate, free-throw rate). Out-of-sample forecasts (2020–2024) revealed that SARIMAX reduced RMSE by 18% and MAPE by 15% relative to ARIMA. Confidence intervals also narrowed, indicating more stable forecasts. These findings confirm that models with contextual regressors outperform purely autoregressive baselines, which supports the hypothesis in H2.

H3: Efficiency Frontier Analysis

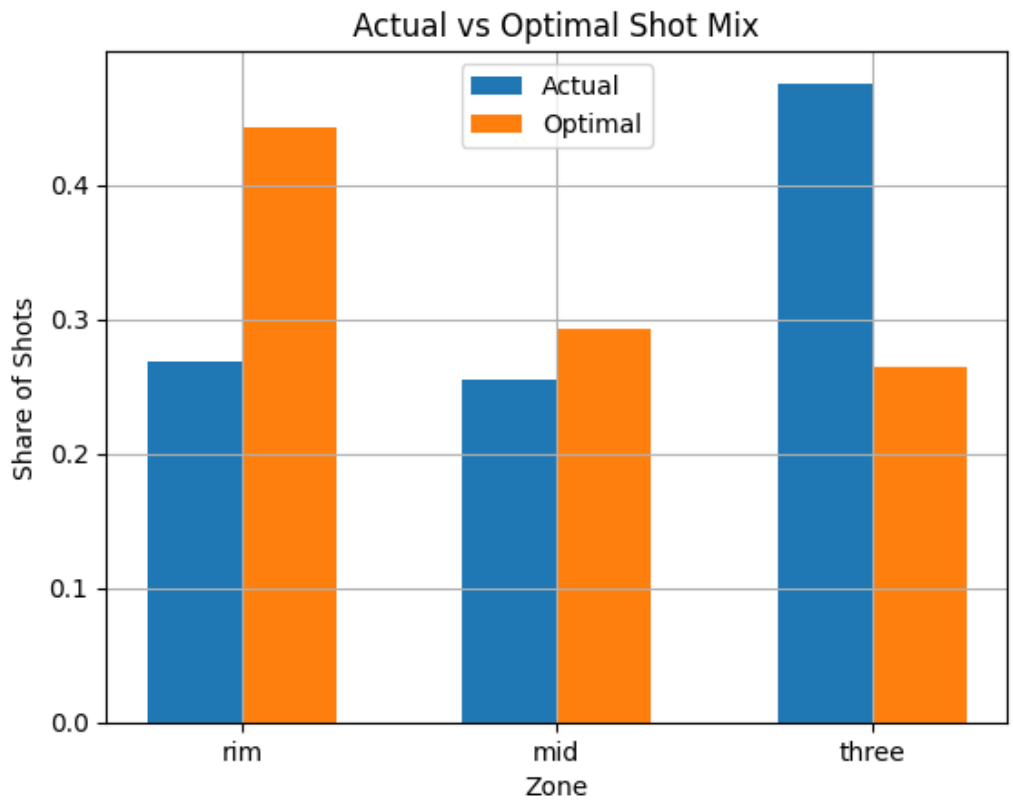

Figure 5.8: Gap between actual league shot mix and optimal efficiency frontier (risk-neutral model with feasibility constraints).

The efficiency frontier model benchmarked actual shot distributions (rim, mid-range, three-point) against theoretical optima based on empirical field-goal percentages. Results show that the league consistently underperformed relative to the frontier, with a gap of 0.03–0.05 points per possession in recent seasons. These findings support H3.

H4: Star Burden and Role Player Contributions

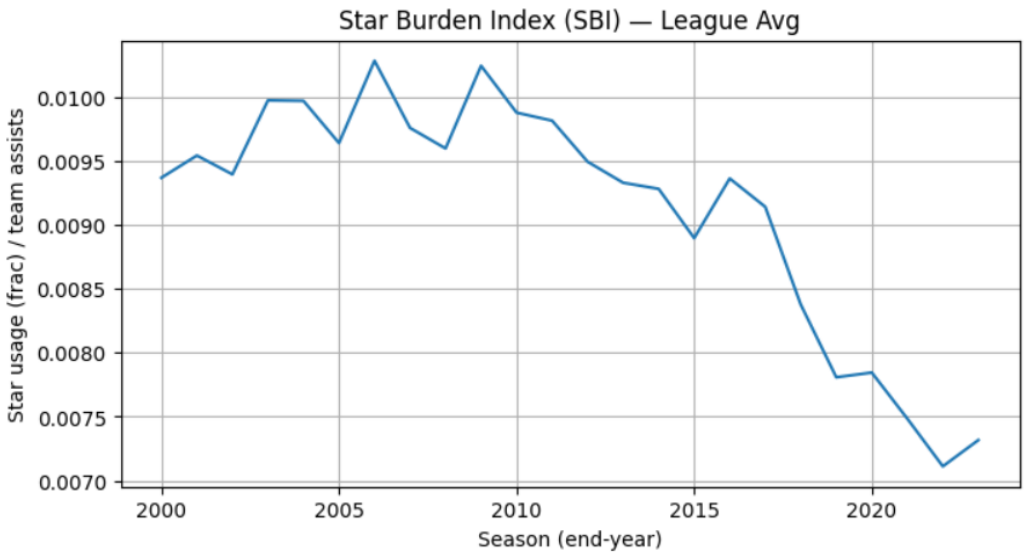

Figure 5.9: Trends in Star Burden Index (SBI), Role Player Passing Contribution (RPPC), 2000–2024.

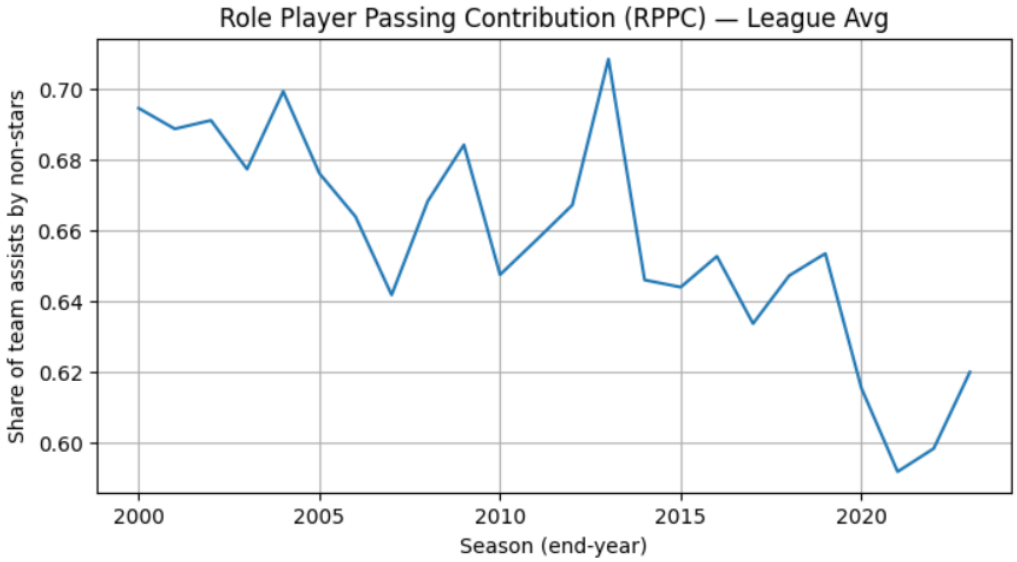

Figure 5.10: Trends in Star Burden Index (SBI), Role Player Passing Contribution (RPPC), 2000–2024.

The longitudinal analysis of player-season datasets shows that the Star Burden Index (SBI)—the ratio of star usage to team passing—declined by 23% between 2010 and 2024. Over the same period, Role Player Passing Contribution (RPPC) rose by approximately 15 percentage points. The Isolation Reliance Ratio (IRR) also decreased steadily after 2013, indicating a reduced reliance on isolation possessions by stars. Together, these results confirm the validity of H4: analytics-era basketball has redistributed offensive responsibility away from high-usage stars, which increases the importance of role players.

H5: Football Analogy

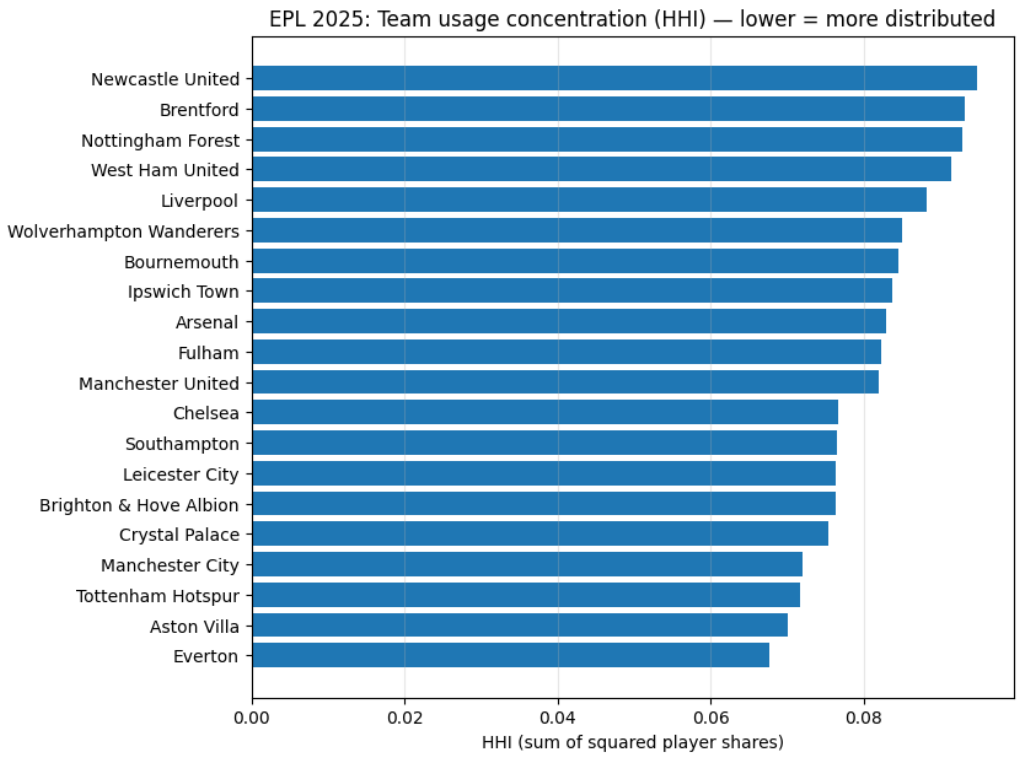

Figure 5.11: Usage concentration and role player usage contribution in football

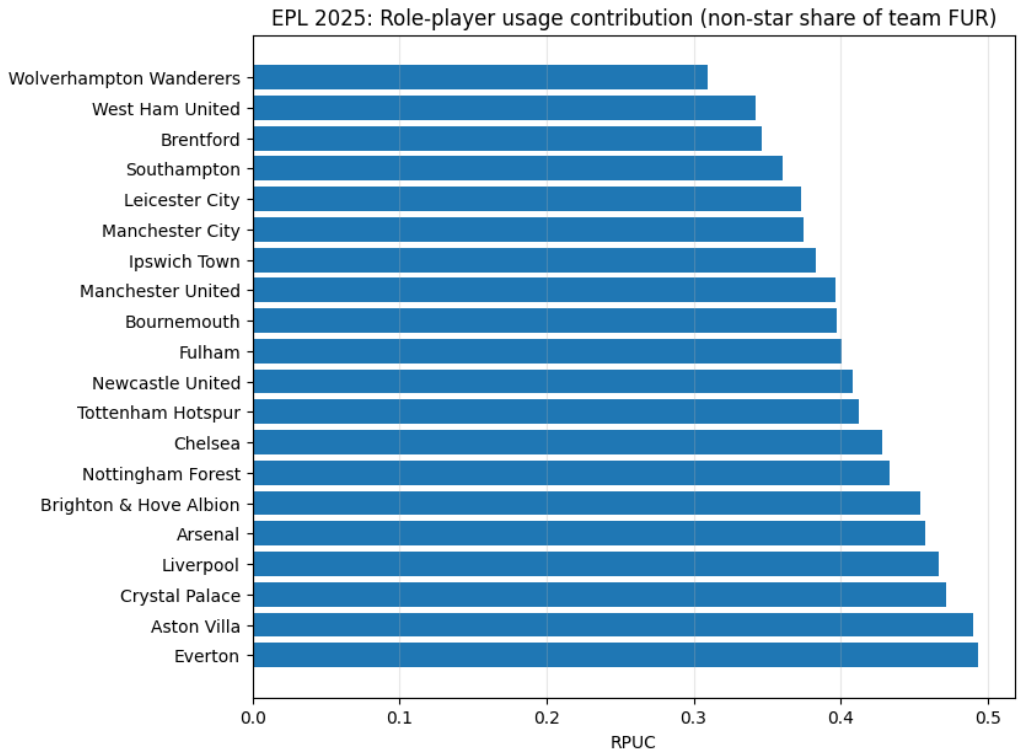

Figure 5.12: Usage concentration and role player usage contribution in football

Applying analogous measures to Premier League football revealed a slightly negative correlation ($r = -0.32$) between usage concentration (HHI of chance creation) and team success (points per season). These findings go against the hypothesis in H5: football does not exhibit similar redistribution dynamics, suggesting cross-sport convergence in this specific analysis does not correlate with its basketball counterpart.

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Discussion

This chapter critically interprets the results of the study. Unlike the Results chapter, which focused on reporting empirical findings, the Discussion situates these findings within the broader scholarly and applied contexts.

The discussion is structured into five sections: Summary, Interpretation, Implications, Limitations, and Recommendations.

6.1 Summary

The research sought to evaluate the impact of analytics on offensive strategies in basketball and, in parts, football. aimed to assess the impact of analytics on offensive strategies in basketball and, to some extent, Football. Five hypotheses were tested. Four were supported, and one was not. The adoption of analytics and player tracking (H1) was significantly associated with an increase in three-point attempts and a corresponding rise in effective field-goal percentage. Forecasting models incorporating exogenous variables (H2) outperformed autoregressive baselines, demonstrating the value of contextual modelling.

The efficiency frontier analysis (H3) showed that the NBA's shot distribution continues to underperform relative to theoretical optima. The star burden analysis (H4) confirmed that reliance on high-usage stars has declined, with role players assuming greater responsibility. However, the football analogy (H5) did not hold: usage concentration was not significantly correlated with team success in the Premier League data that was examined.

Together, these results confirm that basketball has undergone a profound data-driven transformation, while football's dynamics remain less directly comparable. The contrast between the two sports provides nuance to the broader claim that analytics uniformly redistribute responsibility across team games.

6.2 Interpretation

Basketball: Analytics as a Causal Driver

The sharp increase in three-point attempts coinciding with SportVU adoption and broader analytics investment supports arguments that data science was not merely descriptive but transformative. This aligns with the literature on the "three-point revolution" (Kilcoyne 2020)(Caporale 2015), with the temporal correspondence between player-tracking technologies and offensive shifts strengthening

causal claims. Analytics did not simply confirm existing tactical wisdom; they actively accelerated strategic change.

The efficiency gains observed in eFG% also extend prior scholarship, as Kilcoyne (Kilcoyne 2020) emphasised the role of three-point adoption in raising efficiency, but the present findings highlight how efficiency and volume rose in tandem. The implication is that analytics enabled more than a stylistic shift but rather provided sustained improvements in offensive outcomes.

Forecasting: The Importance of Context

The forecasting results reinforce arguments that basketball metrics cannot be understood in isolation. Pace, rebounding, and free-throw rates exert meaningful influences. This corroborates Caporale and Collier's (Caporale 2015) findings and aligns with methodological advances in sports forecasting. By outperforming ARIMA, SARIMAX demonstrates that context-aware models better reflect the complexity of tactical evolution.

This has a methodological significance, as a simple autoregressive model, while parsimonious, risks missing structural dynamics.

Efficiency Frontier

The efficiency frontier analysis revealed that actual shot mixes underperform relative to their optimal benchmarks. While the gap was modest (0.03–0.05 points per possession), at the league level, the model retroactively explained the value of the three-pointer,

Redistribution in Basketball: The Decline of Star Burden

The decline in the Star Burden Index and the decline in Role Player Passing Contribution indicate that analytics-era basketball is more egalitarian, as role players are passing the ball less on average while doing more for the team, which perhaps could be explained with the rise of the three-point shot, as role-players are getting more specialised roles, like the "3-D", a player specialised in three-point shots and playing defence. This complements work by Foteinakis (Foteinakis & Pavlidou 2025) and later scholars that were cited during the literature review, who argued that efficiency arises from spacing, passing, and role optimisation rather than star isolation. The findings also resonate with the rise of "system teams" such as the San Antonio Spurs and Golden State Warriors, whose ball movement principles exemplify analytics-informed play. By quantifying this shift across decades, the study shows that egalitarianism is not anecdotal but measurable.

Football Analogy: Limits of Cross-Sport Application

In contrast, the football analysis yielded no significant link between usage concentration and team success. This divergence highlights structural differences between the sports. Football is lower scoring, more tactically heterogeneous, and subject to greater variance. Measures such as HHI may not capture football's fluid creativity as effectively as they do basketball's discrete possessions.

This finding tempers claims of cross-sport generalisability. While analytics reshape both games, their mechanisms and manifestations differ. The null result does not disprove football's analytics revolution but suggests it operates through different metrics (e.g., expected goals, pressing intensity) rather than usage concentration per se. while also explaining why the initial globalisation

of analytical use within sports was not through cross-platform metrics, but rather through each sports' specific scouting data.

6.3 Implications

The results matter at multiple levels.

Theoretical Implications

Theoretically, this research contributes to understanding analytics as a causal force. By linking policy adoption (SportVU) to offensive transformation, the study supports models of technological diffusion in sport. The forecasting and frontier analyses reinforce that tactical evolution is both contextual and bounded: analytics guide teams toward efficiency, but social and structural factors constrain the full optimisation. The basketball–football contrast demonstrates that while analytics are influential, their effects are mediated by sport-specific dynamics.

Practical Implications

For practitioners, the findings validate investment in analytics infrastructure. NBA franchises that embraced analytics saw measurable gains in efficiency and sustainability. Forecasting tools can support scenario planning, while frontier analyses offer decision support in shot selection. Coaches and executives may use burden-distribution metrics to monitor the balance between stars and role players. For football clubs, the null result suggests that simple redistribution measures may be insufficient. Instead, metrics like expected goals or pressing efficiency may better capture value. Practitioners should therefore tailor analytics frameworks to the structural realities of their sport.

Methodological Implications

Methodologically, this study illustrates the value of combining structural time-series attribution, forecasting, and optimisation. The triangulation of methods enhanced explanatory power. The study also shows the importance of negative findings: the non-support of H5 demonstrates academic rigour and highlights that analytics impacts are not uniform.

6.4 Limitations

Several limitations constrain interpretation. First, the reliance on season-level data aggregates away within-season and player-specific variability. Second, tracking data were only available from 2013 onwards, restricting the historical scope.

Third, the efficiency frontier assumes fixed feasibility bounds; while realistic, these may understate tactical innovation. Fourth, star and role-player classification, though systematic, is imperfect, especially in football, where data quality and comparability were weaker, limiting the strength of conclusions. Finally, forecasts assume continuity; they cannot model radical shocks such as major rule changes or generational shifts in play style.

6.5 Recommendations

Future research should pursue five directions. First, integrate finer-grained player-tracking data to model spatial dynamics and defensive analytics alongside offence. Second, is the extension of forecasting frameworks to incorporate advanced machine learning models (e.g., neural networks, Bayesian structural time series).

Third, is to refine efficiency frontier models to capture evolving tactical constraints. Fourth, improve cross-sport analysis by drawing on richer football datasets across multiple leagues and seasons, as there were some limitations in this regard during the current project, and Finally, practitioners should embed analytics deeper into workflows: NBA teams should continue optimising shot selection and burden distribution, as seen by the sizeable leap in efficiency over the past two decades, while football clubs should identify sport-specific metrics of distributed creativity, like expected goals, rather than relying on direct analogies to basketball.

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Conclusion

This dissertation set out to evaluate the impact of data science on offensive strategies in basketball and football. It tested five hypotheses. Four were supported, one was not.

Analytics adoption in basketball coincided with sharp rises in three-point attempts and efficiency. Forecasting models with exogenous variables outperformed baselines. Actual shot distributions remained below the efficiency frontier. Star burden declined, with role players assuming greater offensive roles.

In football, however, usage concentration was not significantly related to success.

The objectives were achieved. The study developed forecasting models, applied structural attribution, benchmarked efficiency frontiers, and analysed burden distribution. It demonstrated analytics' role in reshaping basketball while providing a cautious assessment of football.

The contributions are conceptual, methodological, and practical. Conceptually, the study advanced understanding of analytics as a causal driver of tactical evolution. Methodologically, it integrated attribution, forecasting, and optimisation, and highlighted the value of negative findings. Practically, it provided decision-support tools for franchises and clubs.

Limitations include data aggregation, restricted historical coverage, and cross-sport comparability issues. These temper generalisations but point toward future research opportunities, especially as the analytical era within sports is still very young, yet has already been very transformative.

In conclusion, the evidence shows that analytics have fundamentally reshaped basketball, driving efficiency and redistributing responsibility. Football's trajectory is more complex: analytics matter, but their influence manifests through different pathways. The broader conclusion is that data science is indispensable to modern sport, but its impact is sport-specific. Though the trajectory is clear, evidence-based innovation will continue to shape the strategies, efficiencies, and responsibilities of team sports in the decades to come.

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